NRES 746, Week 10, Lecture 2: Comparing Pond Models with LOO (take-home)

Name: _____

Due Monday. This uses the 60-pond survey from Lab 8 and the Stan setup from last week's take-home. Fitting all four models takes a few minutes.

Using AI: each coding question is labeled with a recommendation. After each one, circle how you actually used AI.

What to turn in: your filled-in answers and a screenshot of your `loo_compare()` output (upload in WebCampus). You don't need to submit your code.

Part 1: The model

Here is Lab 8's logistic-binomial model in Stan. `X` holds whichever standardized predictors you pass in, and `prior_sd` sets the width of the Normal(0, `prior_sd`) prior on every slope. To compute LOO, Stan must also save the **log-likelihood of each pond** at every draw, in `generated quantities`. One line is missing.

```
library(cmdstanr)
ponds <- read.csv(
  "https://kevintshoemaker.github.io/NRES-746/data/BdRegionalSurvey.csv")
X_all <- scale(as.matrix(ponds[, 2:9]))   # temp, elev, canopy, cond, area,
                                          # fish, bullfrogs, road

pond_code <- "
data {
  int<lower=1> N;
  int<lower=1> K;
  matrix[N, K] X;
  array[N] int<lower=0> n_frogs;
  array[N] int<lower=0> n_pos;
  real<lower=0> prior_sd;
}
parameters {
  real b0;
  vector[K] beta;
}
model {
  b0 ~ normal(0, 2.5);
  beta ~ normal(0, prior_sd);
  n_pos ~ binomial_logit(n_frogs, b0 + X * beta);
}
generated quantities {
  vector[N] log_lik;
  for (i in 1:N)
    log_lik[i] = ____________________________________ ;
}
"
pond_model <- cmdstan_model(write_stan_file(pond_code))

# fits the model with the predictor columns in `cols` and prior SD `s`
fit_ponds <- function(cols, s) {
```

```
  pond_model$sample(
    data = list(N = nrow(ponds), K = length(cols), X = X_all[, cols, drop = FALSE],
                n_frogs = ponds$frogs_swabbed, n_pos = ponds$bd_positive,
                prior_sd = s),
    chains = 4, parallel_chains = 4, seed = 746, refresh = 0)
}
fit_b <- fit_ponds(c(1, 7, 3), 2.5)   # Model B: temp, bullfrogs, canopy
loo_b <- fit_b$loo()
loo_b
```

1a. *(Recommended: no AI.)* Fill in the missing line. (Hint: the `model` block's likelihood, for pond `i` only. Stan's `_lpmf` functions use a bar: `f_lpmf(y | parameters)`.) Then run the code. **Check:** Model B's `looic` should be close to 221.8.

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

1b. Why does LOO need the log-likelihood of each pond separately, rather than one total for all 60?

Part 2: Four models

2a. *(AI optional.)* Fit the four models below with `fit_ponds()` and `$loo()`, and fill in the table. Model D has the same predictors as C but a much narrower prior.

Model	Predictors	prior_sd	elpd_loo	p_loo	looic	max Pareto k
A	temp	2.5				
B	temp, bullfrogs, canopy	2.5				
C	all eight	2.5				
D	all eight	0.3				

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2b. *(AI optional.)* Run `loo::loo_compare(list(A = loo_a, B = loo_b, C = loo_c, D = loo_d))` and record each model's `elpd_diff` and `se_diff` (the best model shows 0 and 0).

Model	elpd_diff	se_diff
A		
B		
C		
D		

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

Part 3: Interpret

3a. Is the best model clearly better than the runner-up? Use `elpd_diff` and `se_diff`.

3b. Models C and D have the same eight predictors. Why is D's `p_loo` (the effective number of parameters) smaller, and why might it predict new ponds better?

3c. Compare the LOO ranking with the AIC table from Lab 8 Act 1. What does LOO tell you that AIC didn't?

Reflection

Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?